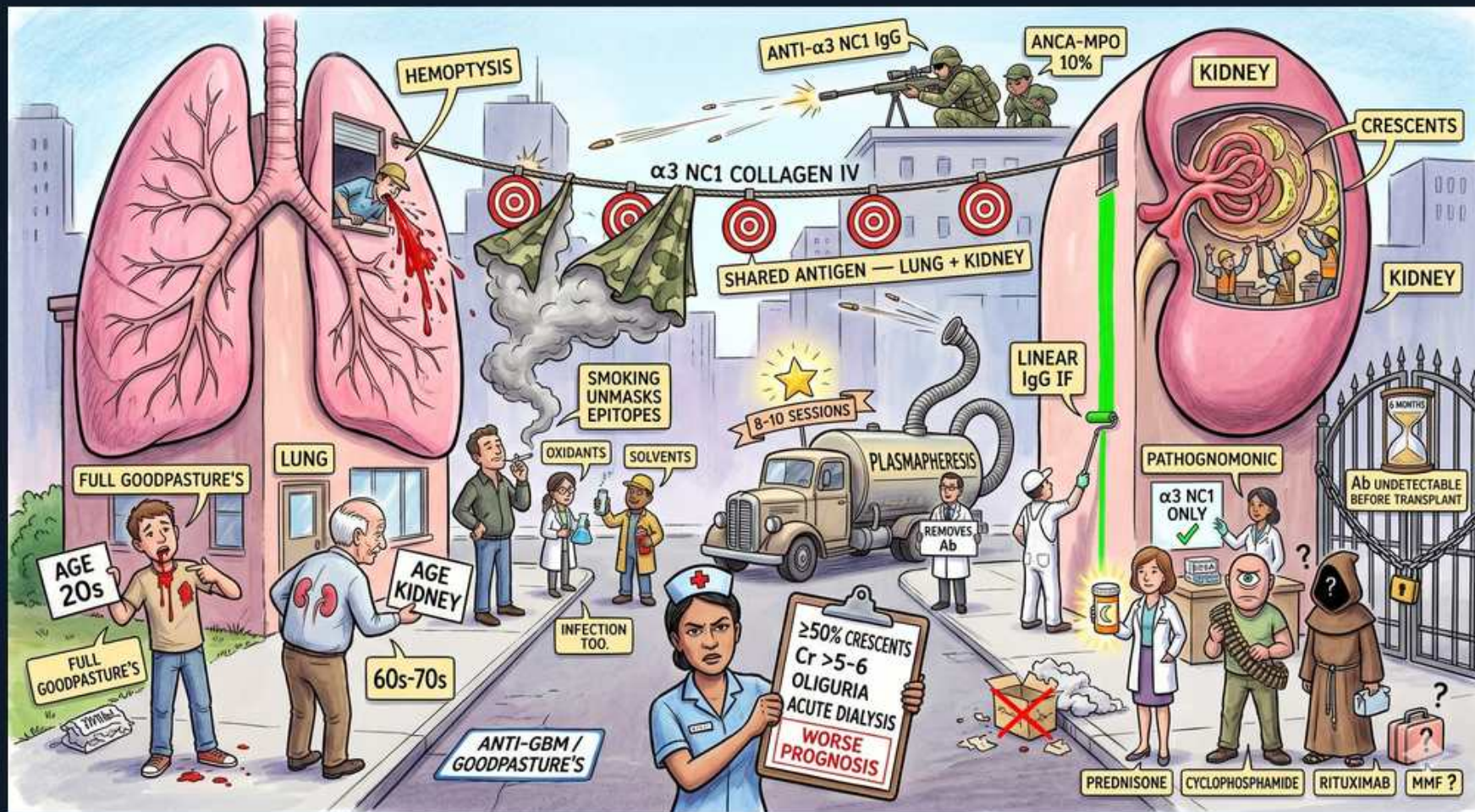


Glomerular (Nephritic) — Anti-GBM / Goodpasture



1. Anti-GBM / Goodpasture

WHAT YOU SEE

A nurse holding a sign reading ANTI-GBM / GOODPASTURE'S — the title card for the lung-plus-kidney antibody disease.

WHAT IT ENCODES

Anti-GBM disease is caused by autoantibodies directed at the glomerular basement membrane, producing a rapidly progressive (crescentic) nephritic glomerulonephritis. When the same antibodies also attack the alveolar basement membrane and cause pulmonary hemorrhage, the syndrome is called Goodpasture's. It is one of the three classic causes of RPGN (pulmonary-renal syndrome), alongside ANCA vasculitis and immune-complex GN. Think 'linear ribbon' on IF as the single highest-yield discriminator.

2. Anti- α 3 NC1 collagen IV

WHAT YOU SEE

Soldiers firing ANTI- α 3 NC1 IgG arrows into red-and-white dartboards labelled α 3 NC1 COLLAGEN IV — the antibody-arrow hitting the collagen bullseye IS the autoantibody striking the α 3-NC1 target antigen.

WHAT IT ENCODES

The antibody (an IgG) targets the noncollagenous (NC1) domain of the α 3 chain of type IV collagen — the 'Goodpasture antigen.' This α 3 epitope is normally hidden within the collagen network and only certain basement membranes (glomerular and alveolar) express it richly, explaining the kidney-lung specificity. Diagnosis is confirmed serologically by anti-GBM antibody ELISA plus the linear IgG biopsy pattern. Mnemonic: 'GBM = Goodpasture, Basement membrane, Mac (type IV) collagen α 3.'

3. Shared lung + kidney antigen

WHAT YOU SEE

A clothesline strung between a lung and a kidney labelled SHARED ANTIGEN — LUNG + KIDNEY — the single line linking both organs shows one antibody hitting the antigen shared by both basement membranes.

WHAT IT ENCODES

The α 3-NC1 collagen IV antigen is shared between alveolar and glomerular basement membranes, so a single autoantibody can strike both organs — the basis of the 'pulmonary-renal syndrome.' This is why a patient can present with both hemoptysis and hematuria/AKI simultaneously. Other classic pulmonary-renal syndromes to keep on the differential are GPA, MPA, and lupus.

4. Hemoptysis / lung

WHAT YOU SEE

A bleeding lung in a window coughing red, labelled HEMOPTYSIS — the blood pouring from alveoli depicts diffuse alveolar hemorrhage.

WHAT IT ENCODES

Diffuse alveolar hemorrhage produces hemoptysis, dyspnea, and bilateral infiltrates; only the lung-plus-kidney form is termed Goodpasture's, whereas kidney-only disease is 'anti-GBM disease.' Lung involvement is much more likely in smokers and younger men and can be life-threatening from massive bleeding. DLCO is paradoxically INCREASED in alveolar hemorrhage because intra-alveolar blood binds inhaled CO — a classic board pearl.

BOARD TRAP

DLCO is decreased in alveolar hemorrhage (it is actually elevated because alveolar blood takes up the CO)

5. Smoking unmasks epitopes

WHAT YOU SEE

Plumes of SMOKE, OXIDANTS and SOLVENTS billowing over the lung — the fumes peeling back the membrane depict inhaled oxidants unmasking the hidden α 3-NC1 epitope.

WHAT IT ENCODES

Inhaled oxidants — cigarette smoke, hydrocarbon solvents, cocaine — injure the alveolar membrane and expose the normally sequestered α 3-NC1 epitope, allowing circulating antibody to bind and trigger pulmonary hemorrhage. This is why pulmonary involvement clusters in smokers while non-smokers more often have renal-limited disease. Counseling smoking cessation is part of management.

6. Linear IgG IF

WHAT YOU SEE

Linear IgG IF green ribbon

WHAT IT ENCODES

Immunofluorescence shows smooth, continuous LINEAR IgG deposition along the entire GBM — the pathognomonic finding because the antibody binds an evenly distributed intrinsic antigen rather than depositing as discrete complexes. This linear 'ribbon' pattern is the single best discriminator from immune-complex GN. The only other classic cause of linear IgG staining is diabetic nephropathy (nonspecific trapping), but there the clinical picture differs entirely.

BOARD TRAP

Granular 'lumpy-bumpy' deposits — that pattern indicates immune-complex GN (PSGN, lupus, endocarditis), not anti-GBM, which is smooth and linear

7. Crescents / RPGN

WHAT YOU SEE

A kidney cutaway with a glomerulus capped by a crescent-shaped wedge of cells, tagged CRESCENTS — the crescent filling Bowman's space IS crescentic RPGN.

WHAT IT ENCODES

Severe GBM injury ruptures capillary walls, allowing fibrin and macrophages to spill into Bowman's space and form cellular crescents — the hallmark of crescentic GN, which clinically manifests as RPGN (creatinine doubling over days to weeks). Anti-GBM is the Type I (anti-GBM antibody) pattern of RPGN; Type II is immune-complex and Type III is pauci-immune/ANCA. Crescents demand urgent biopsy and treatment.

8. >50% crescents = worse

WHAT YOU SEE

>50% crescents, Cr >5–6, oliguria, dialysis sign

WHAT IT ENCODES

Poor renal-prognosis markers are oliguria, creatinine above roughly 5–6 mg/dL (or dialysis-dependence at presentation), and crescents in more than 50% of glomeruli. Patients who are already dialysis-dependent with widespread crescents rarely recover renal function even with aggressive therapy. However, pulmonary hemorrhage is still treated urgently regardless of renal salvageability because it is immediately life-threatening.

9. Normal complement

WHAT YOU SEE

A panel reading PAUCI OF IMMUNE COMPLEXES / α 3 NC1 ONLY — the lack of immune-complex deposits is why complement stays normal (direct antibody binding, not complement-consuming complexes).

WHAT IT ENCODES

Serum complement (C3/C4) is NORMAL in anti-GBM disease because the pathology is direct antibody binding to an intrinsic antigen, not deposition of complement-consuming immune complexes. Contrast with the LOW-complement nephritic causes: PSGN, lupus, endocarditis-associated GN, and MPGN. Normal complement + linear IgG + anti-GBM titer nails the diagnosis.

BOARD TRAP

Low C3 — complement is normal here; low complement instead points to PSGN, lupus, endocarditis, or MPGN

10. Plasmapheresis

WHAT YOU SEE

A PLASMAPHERESIS truck tagged REMOVES Ab / 8-10 SESSIONS hauling plasma away — the truck carting fluid off IS plasma exchange physically removing circulating anti-GBM antibody.

WHAT IT ENCODES

Urgent plasmapheresis (plasma exchange) physically removes circulating anti-GBM antibody and is the cornerstone of induction — typically daily or alternate-day sessions for about 2–3 weeks (roughly 8–14 exchanges) until antibody titers fall and hemorrhage stops. It is the one feature that distinguishes anti-GBM management from most other GN: you must rapidly clear the offending antibody. Start it immediately on strong clinical suspicion, even before biopsy, especially with pulmonary hemorrhage.

11. Immunosuppression

WHAT YOU SEE

A shelf of bottles labelled PREDNISONE, CYCLOPHOSPHAMIDE, RITUXIMAB and MMF — the drug line-up is the immunosuppression that shuts off new antibody production.

WHAT IT ENCODES

Plasmapheresis is paired with immunosuppression to stop new antibody production: high-dose corticosteroids (e.g., IV methylprednisolone pulse 500–1000 mg/day x3, then oral prednisone ~1 mg/kg/day taper) plus cyclophosphamide (oral ~2 mg/kg/day or IV pulses). Rituximab/MMF are used in selected or relapsing cases. The triad — plasmapheresis + steroids + cyclophosphamide — is the high-yield induction regimen. Unlike many GN, relapse is uncommon once antibodies clear, so prolonged maintenance is usually unnecessary.

12. Bimodal age

WHAT YOU SEE

Two patients side by side — a 20-something man (LUNG) and a 60-70s man (KIDNEY) — the two age peaks depict the bimodal distribution and who gets lung vs renal-limited disease.

WHAT IT ENCODES

Anti-GBM disease has a bimodal age distribution: a young-adult peak (20s–30s, more often men) who present with combined lung-plus-kidney Goodpasture's, and an older peak (60s–70s) who more often have renal-limited, isolated kidney disease. Smoking status largely drives whether the lungs are involved. Recognizing the demographic helps anchor a vignette.

13. Antibody clears before transplant

WHAT YOU SEE

An hourglass beside a sign reading Ab UNDETECTABLE BEFORE TRANSPLANT — the waiting-glass means delaying transplant until anti-GBM titers are undetectable to avoid recurrence.

WHAT IT ENCODES

Before kidney transplantation, confirm that anti-GBM antibody titers have been undetectable for at least ~6–12 months to avoid recurrent disease in the allograft. Transplanting too early, while antibody is still circulating, risks immediate recurrence against the new GBM. This is a favorite exam distinction from other GN where timing of transplant is less antibody-driven.